CNC controller engineering

Reset repair and standalone dual-watchdog candidate

30 September 2026 | Source and circuit review | No manufacturing release

1. Corrected FPGA reset buffer - next drawing

U4 now follows Nexperia: NC1 / A2 / GND3 / Y4 / VCC5.
Primary-source checks reject the historical wrong map even when expectations agree.
The corrected circuit is propagated into the connected assembly.
RC timing, startup handoff, partial power and actual FPGA operation remain open.

2. Dual independent watchdog frontends - final drawing

Separate motion and general-output TPS3851 timers; 3.3nF C0G timing capacitors.
Buffered WDI and RESET/WDO qualification; all pins are internal review boundaries.
Standalone only: full permission latches and FPGA self-tests are not integrated.
Required progress pulses differ from the historical endpoint heartbeat toggle.

3. Integration issues identified during review

T20 configuration pull-ups overpower the proposed 100k input pull-downs.
SET1 requires a qualified LOW interface; startup conditioning needs revision.
The 150us SET1 setup is nominal; its maximum and self-test window remain open.
The 17.44ms full-path timing screen uses unmeasured design allowances.

4. Evidence and next executable milestone

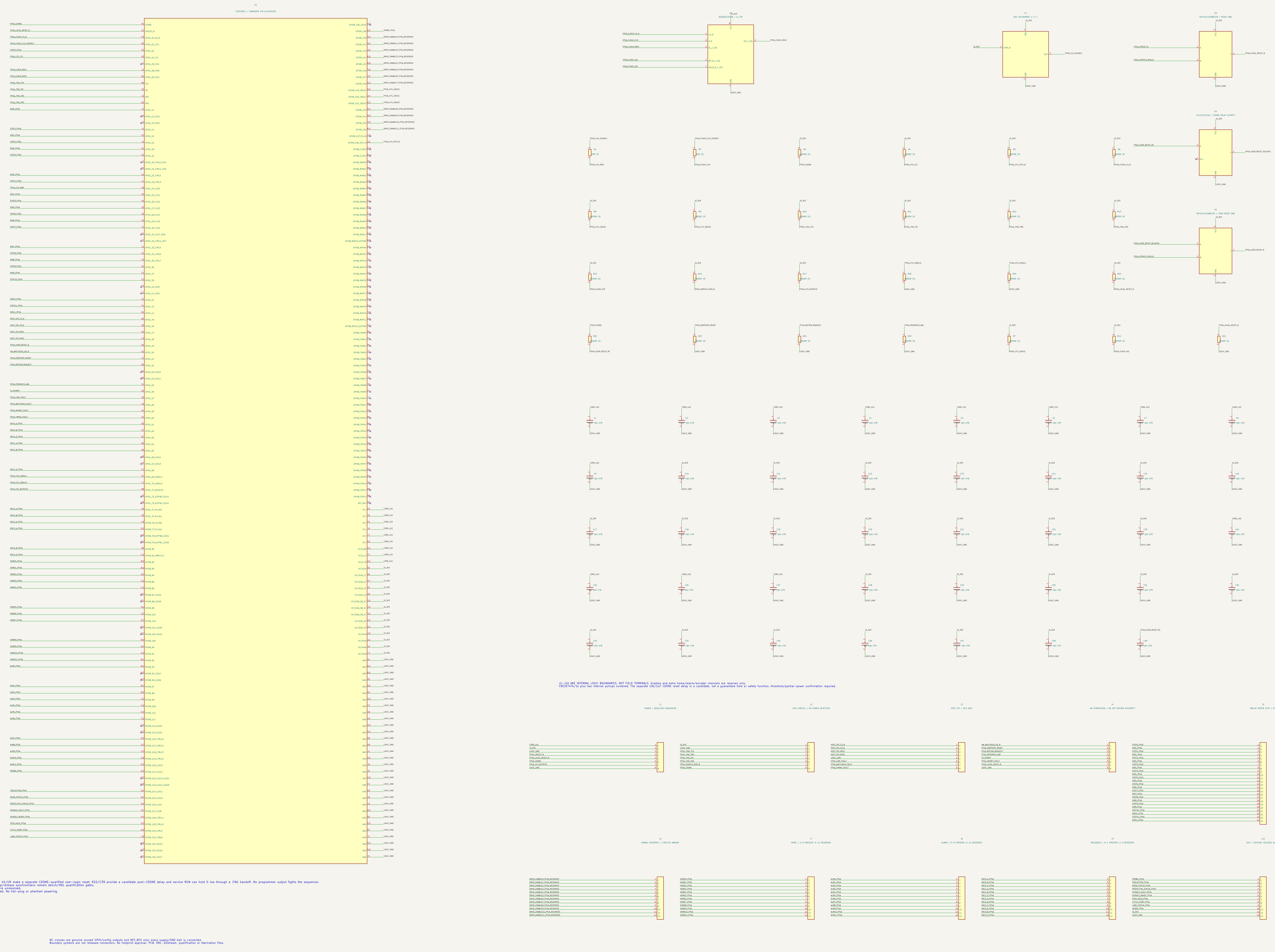
295 connected host tests, 28 fresh native exports and 49 historical HDL checks pass.
Assembly ERC: zero errors; 22 documented floating-substrate warnings remain.
Next: actual eight/twelve-channel CNC3 / IS-1 / LinuxCNC disconnected integration.
Power and whole-package cost closure retain required I/O, diagnostics and repair.

Source receipts: reports/next-implementation-20260930/README.md
Native schematic pages preserve their original paper sizes; zoom to inspect pins.

T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12–CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20–PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCCA=1.2V. No 5V–tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball; bulk per physical supply group. Decoupling is a layout–study allowance, not a PDN proof.
Active SPI x1 straps111; CSI/TEST high. Flash bus reserved from user RTL. CCK damping33R is provisional.
CDONE is open–drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not Y1.


U1
T20F256I4 / CANDIDATE PIN ALLOCATION
U2
W25Q32JVSSIQ / x1 SPI
Y1
ASE–50.000MHZ–L–C–T
U3
SN74LVC1G08DCKR / RESET AND
U4
74LVC1G17GW / CDONE DELAY SCHMITT
U5
SN74LVC1G08DCKR / USER RESET AND
J1
POWER / QUALIFIED SEQUENCER
J2
JTAG SERVICE / NO POWER INJECTION
J3
HOST SPI / 3V3 ONLY
J4
HW PERMISSION / NO ESP MOTION AUTHORITY
J5
TWELVE MOTOR STEP / DIR
J6
ENABLE RESERVES / CIRCUITS ABSENT
J7
HOME / 0–5 PRESENT: 6–11 RESERVED
J8
ALARM / 0–5 PRESENT: 6–11 RESERVED
J9
ENCODERS / 0–1 PRESENT: 2–3 RESERVED
J10
AUX / EXISTING ISOLATED INPUTS
#FLG01
SUPPLIED BY UPSTREAM
#FLG03
SUPPLIED BY UPSTREAM
CORE_1V2
IO_3V3
LOGIC_GND

J1–J10 ARE INTERNAL LOGIC BOUNDARIES, NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESET47k/1k plus two internal pullups screened. The separate 10k/1uF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial–power confirmation required.

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE–qualified user–logic reset; R22/C39 provide a candidate post–CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User–logic reset release threshold, RC tolerance, discharge time, tUSER handoff and startup/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary: actual keyed connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate reset cycle, valid rails and machine outputs inhibited. No hot–plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bitstream, qualification or fabrication files.

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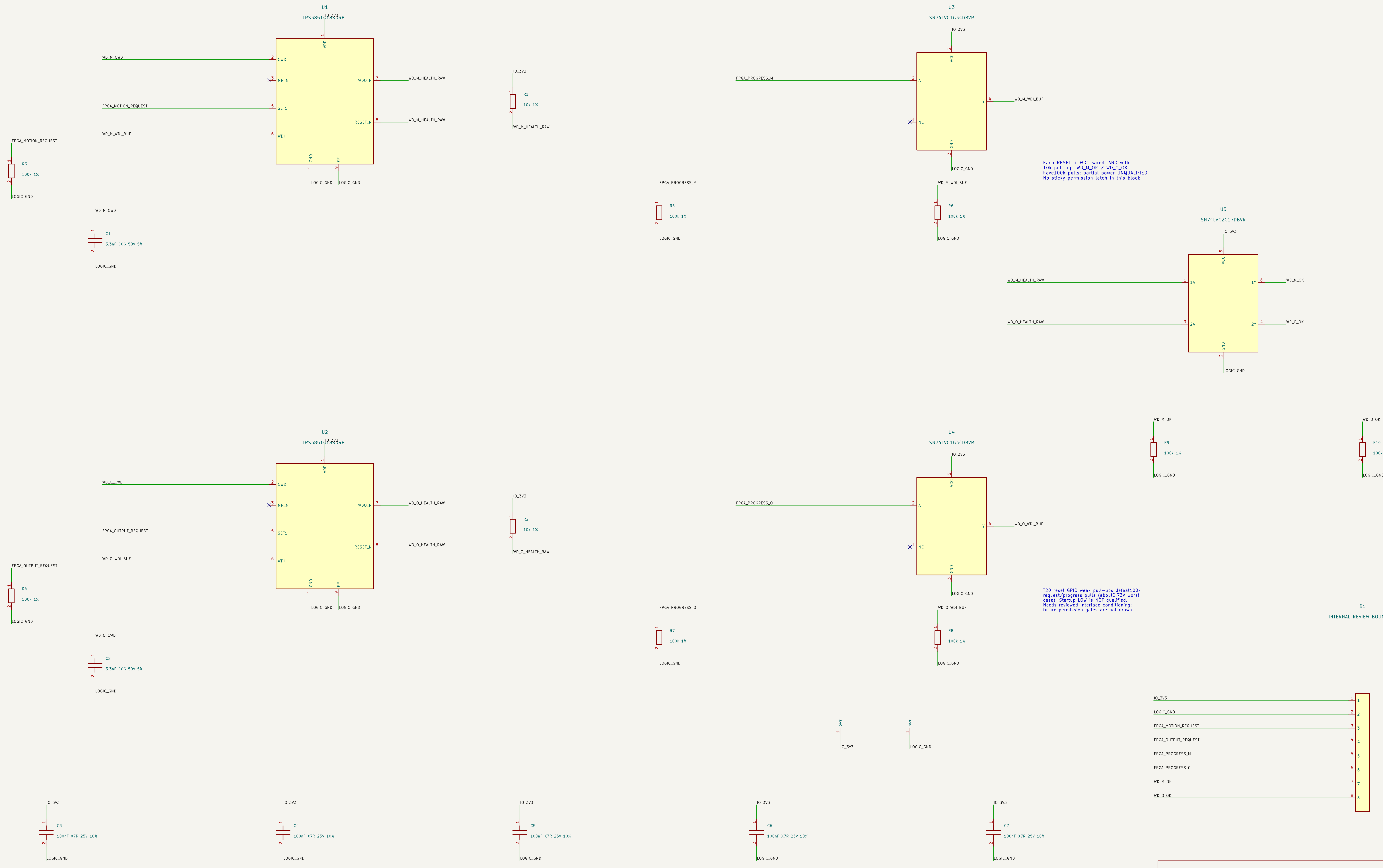
IS–1 DUAL WATCHDOG TIMER FRONTENDS – STANDALONE REVIEW CANDIDATE

NOT CONNECTED TO THE ASSEMBLY. No complete permission chain, latches, field outputs, PCB or physical qualification.

Separate timers / shared IO_3V3 and LOGIC_GND / G18 threshold is not a 3.3V power–good qualification / all inputs INTERNAL logic.

WD_M: motion ARM / UPDATE progress; WD_O: full–I/O progress. SET1=0 disables its timer. MR open per frozen IS–1.



WDI: one 1us high pulse, idle LOW. FALLING edges feed timer.
Old toggle–per–UPDATE gateware is INCOMPATIBLE.
3.3nF C0G: calculated timing, never a measured20ms proof.

Each RESET + WDO wired–AND with
10k pull–up. WD_M_OK / WD_O_OK
have100k pulls; partial power UNQUALIFIED.
No sticky permission latch in this block.

T20 reset GPIO weak pull–ups defeat100k
request/progress pulls (about2.73V worst
case). Startup LOW is NOT qualified.
Needs reviewed interface conditioning;
future permission gates are not drawn.

DRB copper–pad candidate only: paste/stencil, thermal vias, CWD guarding/cleanliness/leakage, routing, partial power and brownout remain unqualified.

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File: is1_watchdogs.kicad_sch
Title: IS–1 dual watchdogs – review only
Size: A2 Date: 2026–09–30 **Rev: r0.1**
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